Topological Decomposition of E_8 Boundary Phase Space and Structural Constraints on the Fine-Structure Constant

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Abstract

We propose a topological framework for the infrared fine-structure constant $\alpha^{-1}(0)$ based on the boundary kinematics of an E_8 gauge field theory compactified on a Complete Intersection Calabi-Yau (CICY) manifold. We decompose the inverse coupling into a primary $U(2)$ principal bundle volume $V_{\text{main}} = 4\pi^3$, a Hodge-orthogonal boundary projection $V_{\text{boundary}} = \pi^2 + \pi$, and a sub-leading spectral correction $\Delta\alpha^{-1} = \delta_{G_2} \cdot (2 + \sqrt{5})$ associated with the non-associative geometry of octonionic G_2-automorphisms.

To establish the structural consistency of the low-energy effective field theory, we evaluate the gauge sector compactification via two topological conditions:

1. **The Equivariant Lift Gate (16-Fermion Sector):** Under the vanishing conditions $H^0 = H^2 = H^3 = 0$ on the twisted bundle $V \otimes L$, the equivariant Lefschetz index locks the chiral spectrum to exactly 3 generations across the character sectors of a discrete $\mathbb{Z}_3 \times \mathbb{Z}_3$ isometry.
2. **The Λ^2-Cohomology Gate (10-Higgs Sector):** Because the global Euler characteristic vanishes ($\chi(\widetilde{X}, \Lambda^2 V) = 0$), we apply the Koszul resolution on the ambient product space $\mathbb{P}^2 \times \mathbb{P}^2$ via the Bott-Borel-Weil theorem. We show that Wilson-line symmetry breaking induces a rank defect in the coboundary matrix M_s, projecting out mass-free color triplets ($m_{\text{triplet}} = 0$) while preserving a single light electroweak Higgs pair ($m_{\text{doublet}} = 1$).

Furthermore, we utilize the Lean 4 interactive theorem prover (using Mathlib4) as an automated type- and charge-consistency checker. Lean 4 formally verifies the internal arithmetic constraints, the $\phi^3 = 2 + \sqrt{5}$ golden ratio identity, and the exact charge-conservation laws on the E_8 weight lattice. The complete differential-geometric stack is formulated as an integrative, structurally constrained framework for precision coupling unification.

Epistemological Declaration (Three-Tier Governance Hierarchy):

Tier 1 (Lean 4 Verified Arithmetic): Machine-checked proof of internal numerical consistency, real algebraic identities ($\phi^3 = 2 + \sqrt{5}$), and charge-conservation arithmetic on the 8D weight lattice.

Tier 2 (Mathematical Structural Proofs): Application of the Koszul sequence, Bott-Borel-Weil evaluation, and index theorems constraining zero-mode spectra on CICY manifolds.

Tier 3 (Physical Model Framework): Phenomenological identification of V_{main} with $U(2)$ phase space and the Dunkl-operator formulation for δ_{G_2}, serving as a predictive proposal subject to ongoing symbolic derivation.

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1 Introduction

The derivation of fundamental dimensionless parameters in quantum field theory remains an open challenge. In this work, we structure the infrared coupling constant $\alpha^{-1}(0)$ via topological boundary conditions of exceptional Lie groups...

2 The Gauge Sector Compactification

2.1 The 16-Fermion Sector and the Equivariant Lift Gate

To determine the net chiral generation count, we analyze the twisted vector bundle $V \otimes L$ over the Calabi-Yau cover $\widetilde{X}$...

2.2 The 10-Higgs Sector and the Λ^2-Cohomology Gate

Because $\chi(\widetilde{X}, \Lambda^2 V) = 0$, global index theorems cannot resolve the presence of mass-free Higgs doublets. We resolve the cohomology groups $H^1(\widetilde{X}, \Lambda^2 V)$ via the short exact Koszul sequence:

$$0 \to \Lambda^2 \mathcal{V} \otimes E^* \xrightarrow{\cdot s} \Lambda^2 \mathcal{V} \xrightarrow{\text{res}} i_*(\Lambda^2 V) \to 0 \tag{1}$$

Evaluating the induced coboundary map δ_s via the Bott-Borel-Weil theorem on the ambient space $\mathbb{P}^2 \times \mathbb{P}^2$ yields the exact multiplet spectrum...

3 Formal Verification in Lean 4

The Lean 4 proof assistant is employed at Tier 1 to verify algebraic sub-routines...

4 Conclusion

We have presented a multi-tiered framework for E_8 boundary kinematics...

References

[1] A. Sommerfeld, *Zur Quantentheorie der Spektrallinien*, Annalen der Physik **356** (1916), 1–94.

[2] The Mathlib Community, *The Lean 4 Mathematical Library (Mathlib4)*, https://github.com/leanprover-community/mathlib4 (2023).